

POTHOLE AND SPEED BREAKER DETECTION

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Abstract

Reliable detection of road-surface anomalies specifically potholes and speed breakers is essential for roadway safety and maintenance planning, yet many existing systems degrade under changes in illumination, weather, and viewpoint. This paper presents an end-to-end computer vision pipeline that combines lighting-aware pre-processing and bounding-box preserving augmentation with convolutional detectors including You Only Look Once (YOLO) version 8 and version 10, Faster Region-based Convolutional Neural Network (Faster R-CNN), and Single Shot MultiBox Detector (SSD). A unified annotation interface supporting conversion between YOLO and Pascal Visual Object Classes (VOC) formats enables cross-framework training, while input resizing is performed within the training stack to preserve label fidelity. The study follows a two-stage design: (i) a 10-epoch baseline comparison across detector families and (ii) focused hyperparameter tuning of the strongest baseline. Models are evaluated using mean Average Precision (mAP) at Intersection over Union (IoU) thresholds from 0.50 to 0.95 (mAP@0.5:0.95), precision, recall, and F1-score across daytime and nighttime conditions. The tuned SSD300 with Visual Geometry Group 16 (VGG16) backbone configuration achieves the best validation performance with mAP@0.5:0.95 up to 0.97, improving precision while maintaining high recall. Error analysis indicates remaining challenges with nighttime glare and very small or oblique targets. Overall, the results support the feasibility of the approach for Advanced Driver Assistance Systems (ADAS) integration and Geographic Information System (GIS) marking, and suggest that further gains will come from expanding data diversity and coverage.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADAS	Advanced Driver Assistance Systems
Adam	Adaptive Moment Estimation (Optimizer)
AdamW	Adam with Weight Decay
AMP	Automatic Mixed Precision
AP	Average Precision
bbox	Bounding Box
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CPU	Central Processing Unit
CSPDarknet	Cross Stage Partial Darknet
CUDA	Compute Unified Device Architecture
DL	Deep Learning
F1	F1-Score (Harmonic Mean of Precision and Recall)
FN	False Negative
FP	False Positive
FPN	Feature Pyramid Network
GFLOPs	Giga Floating Point Operations per Second
GIS	Geographic Information System
GLCM	Gray-Level Co-occurrence Matrix
GPU	Graphics Processing Unit
IoT	Internet of Things
IoU	Intersection over Union
JSON	JavaScript Object Notation
LiDAR	Light Detection and Ranging
LR	Learning Rate
mAP	mean Average Precision
mAP@0.5	mean Average Precision at IoU threshold 0.50
mAP@0.5:0.95	mean Average Precision at IoU thresholds from 0.50 to 0.95
ML	Machine Learning
MobileNetV3	Mobile Network Version 3
NMS	Non-Maximum Suppression
P	Precision
R	Recall
R-CNN	Region-based Convolutional Neural Network
ResNet	Residual Network
RNN	Recurrent Neural Network
ROI	Region of Interest

RPN	Region Proposal Network
SGD	Stochastic Gradient Descent
SRGANs	Super-Resolution Generative Adversarial Networks
SSD	Single Shot MultiBox Detector
SSD300	Single Shot MultiBox Detector with 300×300 input
SVM	Support Vector Machine
TensorRT	NVIDIA TensorRT (Inference Optimizer)
TP	True Positive
Val	Validation
VGG	Visual Geometry Group
VGG16	Visual Geometry Group 16-layer Network
VOC	Visual Object Classes (Pascal VOC)
WD	Weight Decay
YOLO	You Only Look Once
YOLOv7	You Only Look Once Version 7
YOLOv8	You Only Look Once Version 8
YOLOv8m	You Only Look Once Version 8 Medium
YOLOv10	You Only Look Once Version 10
YOLOv10m	You Only Look Once Version 10 Medium
Abbreviation	Full Form
ADAS	Advanced Driver Assistance Systems
AP	Average Precision
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNNs	Convolutional Neural Networks
COCO	Common Objects in Context
CPU	Central Processing Unit
DL	Deep Learning
F1	F1-Score (Harmonic Mean of Precision and Recall)
FN	False Negative
FP	False Positive
FPN	Feature Pyramid Network
GIS	Geographic Information System
GLCM	Gray-Level Co-occurrence Matrix
GPU	Graphics Processing Unit
IoU	Intersection over Union
LR	Learning Rate
mAP	Mean Average Precision
mAP@0.5:0.95	Mean Average Precision averaged over IoU thresholds from 0.50 to 0.95 (COCO standard)

ML	Machine Learning
NMS	Non-Maximum Suppression
R-CNN	Region-Based Convolutional Neural Network
RNN	Recurrent Neural Network
SRGANs	Super-Resolution Generative Adversarial Networks
SSD	Single Shot MultiBox Detector
TP	True Positive
VOC	Visual Object Classes (Pascal VOC)
YOLO	You Only Look Once
YOLOv7	You Only Look Once Version 7
YOLOv8	You Only Look Once Version 8
YOLOv10	You Only Look Once Version 10

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CHAPTER 1: INTRODUCTION

1.1 Background

Potholes and improperly designed speed breakers pose significant challenges to road safety and infrastructure maintenance, particularly in countries like India. In 2022, potholes were responsible for over 4,400 accidents across the nation. (India: road accidents due to potholes | Statista, 2025). These hazardous road conditions not only lead to accidents but also escalate vehicle maintenance costs, fuel consumption, and travel times, imposing substantial economic burdens.

Speed breakers are intended to regulate vehicular speed, but can become perilous when poorly designed. Studies have shown that non-compliant speed breakers often lead to abrupt vehicle movements, loss of control, and increased wear and tear, making them hazardous. (Kumar et al., n.d.).

The economic ramifications of potholes and substandard speed breakers are profound. Damaged roads contribute to higher fuel consumption, increased vehicle repair costs, and disrupted logistics. Traffic congestion resulting from these defects further hampers transportation efficiency, adversely affecting both urban and rural economies. Addressing these challenges necessitates innovative approaches to road maintenance and safety.

Advancements in machine learning and computer vision offer promising solutions for detecting and assessing road anomalies. AI-driven systems can analyse images, sensor data, and real-time traffic information to identify potholes and speed breakers with high precision. For instance, recent research has demonstrated the effectiveness of deep learning models, such as YOLOv8, in accurately detecting road hazards, including potholes and manholes, under various conditions (Khare et al., 2023)

Additionally, the integration of Super-Resolution Generative Adversarial Networks (SRGANs) with object detection models like YOLOv7 has shown improved accuracy in identifying potholes, even in low-resolution images.

As urbanization intensifies and vehicle numbers surge, integrating intelligent road monitoring solutions into existing transportation frameworks becomes imperative. Machine learning-based detection methods offer a cost-effective and scalable approach to identifying hazardous road

conditions, ultimately enhancing safety, improving traffic flow, and preserving infrastructure integrity.

1.2 Problem Statement

Road anomalies like potholes and speed bumps present significant risks to road safety and can cause vehicle damage, underscoring the need for automated detection systems, particularly given the inefficiency of traditional methods like manual assessment. The accurate detection of these anomalies is complicated by several factors such as varying sizes and shapes of potholes, uneven speed breaker construction, lack of adherence to government norms in speed breaker design, and absence of proper markings or cautionary signs. Adverse environmental conditions, including rain, poor lighting, varying weather, distance, and speed, add to the challenge. Limitations in camera quality, sensor accuracy, and computational capabilities further impact the reliability of real-time detection, especially considering street images filled with redundant information, substantial noise, and contamination. To overcome these limitations, this research focuses on leveraging machine learning and computer vision algorithms to enhance detection accuracy and reliability, effectively detecting speed breakers and potholes from different camera angles as well as under diverse environmental conditions.

1.3 Aims & Objectives

The main aim of this research is to develop a robust & efficient model for the detection of speed breakers & potholes using machine learning & computer vision techniques, ensuring reliable performance across various weather conditions and lighting scenarios. The goal of this research is to contribute to effective speed breaker & pothole detection which can be utilized as a robust model in ADAS (Autonomous Driving Assistant System) or for GPS marking of potholes & speed-breakers.

The research objectives are formulated based on the aim of the study, which are as follows:

- To implement pre-processing techniques to enhance image quality, reduce noise & handle different lighting/ weather conditions.
- To design and implement machine learning, computer vision models and other deep learning architectures, for accurate detection and classification of speed breakers and potholes.
- To evaluate performance under various adverse environmental conditions (varying lighting, weather, and road surface).

- To identify the optimal ML/DL model(s) that provide the best balance of accuracy, recall & precision.

1.4 Significance of the Study

This research has the potential to greatly contribute to safer roads & lesser injuries/ fatalities to the public with economic benefits at national and international levels. Key aspects highlighting the importance and potential implications of this study include:

- **Reducing Accidents:** Systems detecting potholes and unmarked speed bumps can give drivers timely warnings so they can avoid dangers, which lowers the number of accidents and injuries.
- **Keeping Vehicles in Good Repair:** Systems enabling drivers to avoid road flaws can lower the cost of vehicle repairs and extend the life of the vehicle.
- **Autonomous Vehicle Advancement:** Autonomous Driving Assistant System (ADAS) cars can accurately detect road anomalies to be safer and more reliable.
- **Improved Road Safety:** Deployment on a large-scale the system has the potential to monitor road conditions and alert drivers of potential hazards, which can reduce accidents and injuries.
- **Economic Advantages:** Lowering vehicle maintenance expenses, enhancing road maintenance effectiveness, and cutting down on accidents can result in considerable financial gains for people, businesses, and governments.
- **Worldwide Applicability:** The research techniques and concepts can be used on road networks around the world, which will improve transportation systems' safety and efficiency.
- **Smart City Initiatives:** Integrating pothole and speed bump detection systems into smart city projects can give city officials and residents real-time data on road conditions.
- **Technology to Assist:** By offering audio alerts regarding road conditions, the technology can be tailored to help people who are blind.
- **Guiding Policy Makers:** The study can help governments and decision-makers spot unauthorized or poorly constructed speed bumps, which will lower the number of accidents and the related financial losses brought on by subpar road infrastructure.

1.5 Scope of the Study

There are following items which are in-scope under this research plan.

- Image Data: Focus on identifying these road anomalies using image data.
- Adverse Environmental Conditions: Specifically address detection challenges posed by varying lighting, and road surface conditions.
- Developing Machine Learning (ML) and Deep Learning (DL) Models: Development, comparison, and evaluation of different ML/DL models.
- Performance Metrics: Evaluation using metrics like accuracy, precision, recall, F1-score, and processing time.

There are following items which are Out-of-Scope under this research plan.

- Sensor Data: Excluding the integration of data from other sensors such as accelerometers, LiDAR, or radar
- GPS Integration: Excluding the use of GPS data for location mapping or real-time driver alerts.
- Real-Time Implementation: Avoiding real-time implementation in a vehicle or mobile device.
- 3D Reconstruction: Excluding 3D pavement reconstruction methods that employ laser scanning or stereo vision.
- Vibration-Based Methods: Not using vibration-based approaches that rely on acceleration sensors.
- ADAS or Autonomous Vehicle Integration: Not integrating the system directly into Advanced Driver Assistance Systems (ADAS) or autonomous vehicle platforms.
- Implementation on specific devices: Not implementing the model on a mobile phone

1.6 Structure of the Study

The structure of the study is as follow:

- Chapter 1 discusses the introduction to the research paper which further contains sub-sections like
 - 1.1 Background,
 - 1.2 Problem Statement
 - 1.3 Aims & Objectives
 - 1.4 Significance of the Study
 - 1.5 Scope of the Study.

- Chapter 2 discusses the related work done in the industry to identify speed breakers & potholes. Initial research started with sensor-based researches which utilises accelerometers in smartphone to identify vibrations to identify potholes & speed breakers. With the advancement in technology, the researches moved a step further towards image based detection. Post 2015, there was a transition to machine learning & deep learning based techniques & using GPS to directly mark these road anomalies into the maps.
- Chapter 3 - Research Methodology - details the proposed methodology adopted in this study. It describes the **dataset** (Kaggle-sourced images with potholes and speed breakers captured under different lighting and weather conditions), **data mining and label organization**, and the use of a **unified annotation interface** to convert between YOLO and Pascal VOC formats. The chapter then presents the **pre-processing pipeline** (including lighting-aware enhancement with CLAHE), **bounding-box-aware data augmentation**, and the **train/validation/test splitting strategy**. It further explains the **modelling techniques** considered (YOLOv8/YOLOv10, Faster R-CNN with MobileNetV3-FPN, and SSD300-VGG16), followed by the **training protocol** (transfer learning, fine-tuning, checkpointing) and the design of **Experiment 1 (baseline comparison)** and **Experiment 2 (SSD hyperparameter tuning)**. The **evaluation metrics** (Precision, Recall, F1, IoU, mAP@0.5:0.95) and the **hardware/software environment** used for training and experimentation are also described.
- Chapter 4 - Results and Discussion - presents and interprets the experimental results. It first reports the outcomes of **Experiment 1**, comparing YOLO, Faster R-CNN, and SSD under a common 10-epoch baseline and highlighting key observations such as the relative strengths of each model and the challenges in night-time scenes. It then discusses **Experiment 2**, where SSD300-VGG16 is further optimized through hyperparameter tuning, and shows how mAP@0.5:0.95, Precision, Recall, and F1 improve over the baseline. Comparative tables, graphs, and qualitative visualizations (prediction overlays) are used to analyze performance across conditions. The chapter also includes an **error analysis**, discussing typical failure cases (e.g., reflective markings, very small potholes, oblique viewpoints) and their implications for real-world deployment.

- Chapter 5 - Conclusion and Future Work - summarizes the study's main findings and contributions. It revisits the research objectives and explains how they were addressed, emphasizing the selection of **SSD300-VGG16 with a conservative configuration** as the recommended model for pothole and speed breaker detection on the given dataset. The chapter then outlines the **limitations** of the current work, such as dataset size and diversity, and proposes **future work**, including expanding the dataset with more diverse roads and marked speed breakers, exploring domain adaptation and semi-supervised learning to improve generalization, and optimizing the final model for real-time deployment (e.g., via ONNX/TensorRT, pruning, and quantization). Possible extensions to video-based detection with temporal

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The challenge of reliably detecting road anomalies such as potholes and speed breakers has been a persistent concern in transportation safety and infrastructure management. Early research on road anomaly detection primarily utilized basic techniques such as threshold methods and manual analysis of sensor data. These initial approaches often involved visual inspections or simple vibration-based methods, which were limited by their inefficiency and subjectivity. The need for more efficient and automated systems became apparent as technology advanced. This section details the progression, starting from sensor-based approaches to image processing techniques and finally to machine learning & deep learning methodologies. However, it's important to note that earlier research often lacked robust accuracy when faced with challenging environmental conditions such as nighttime, fog, mist, and rain. The primary objective of examining this body of work is to gain a comprehensive understanding of previous research and to leverage these insights in the current research endeavour.

2.2 Early Research and Sensor-Based Approaches

Early detection methods for road anomalies relied on vibration-based systems, primarily using accelerometers. (Vittorio et al., 2014) measured vehicle vibrations and compared them against predetermined thresholds to identify potential potholes or speed breakers. This approach provides real-time data about road conditions to facilitate road repairs. This method was cost-effective but contained high false-positive rates. (Mohan et al., 2008) utilized anomaly detection thresholds with accelerometer data. (Perttunen et al., 2011) applied machine learning algorithms to accelerometer data for improved accuracy. (Mednis et al., 2011) introduced a real-time pothole detection system using Android smartphones and accelerometer data. This approach detected sudden vertical accelerations indicating potholes. (Bhoraskar et al., 2012) also used accelerometers to detect road anomalies. (Douangphachanh and Oneyama, 2014) refined these approaches by introducing road roughness classification. (Ramsha Suhail, 2020) proposed the use of piezo-electric sensors attached to the shock absorbers of the front wheels to detect voltage changes caused by road anomalies. While sensor-based methods were effective in detecting basic road anomalies, they were prone to false positives and lacked

robustness in variable road conditions. Moreover, they offered limited spatial coverage due to the dependency on individual users.

2.3 Transition to Image Processing Techniques

To overcome the limitations of sensor-based models, researchers turned to image processing and computer vision techniques. These methods analyzed images or video frames captured by vehicle-mounted cameras or drones to detect road anomalies. Researchers started creating algorithms to spot potholes and speed breakers in road images (Devapriya et al., 2016). (He et al., 2011) introduced a model using two CCD cameras and LED linear light to detect the 3-dimensional cross-section of pavement potholes. (Devapriya et al., 2016) used Gaussian filtering and median filtering to eliminate noise in images to detect speed bumps.

2.4 Transition to Machine Learning & Deep Learning Approaches

The integration of machine learning algorithms marked a significant step toward improving the accuracy and efficiency of road anomaly detection (Gaisser and Jonker, 2017). (Jain et al., 2012) used machine learning for bump detection using accelerometers in various vehicles in 2012. SVM models were employed for pothole detection in pavement images (He et al., 2011). (Bharathi et al., n.d.) proposed a speed breaker identification method using gray-level co-occurrence matrix (GLCM) features. In 2023, (Billah and Chattaraj, 2021) proposed adaptively choosing deep learning models like ResNet50, EfficientNet B7, VGG16 and AlexNet using image processing to detect road conditions. (Hasan et al., 2020) proposed using CNN models to detect speed breakers, potholes in images, recorded video as well as in real-time video.

3D reconstruction methods, leveraging 3D laser scanning and stereo vision, emerged, enabling the creation of detailed road surface models (Kim et al., 2022). (Li et al., 2009) presented a real-time 3D laser scanning system for detecting pavement rutting, shoving, and potholes. (Hou et al., 2007) proposed a stereo-vision-based surface model for better pavement conditioning. (Chang et al., 2005) introduced a clustering approach to quantify the severity and coverage of potholes using 3D point clouds. However, the high costs and computational demands associated with these methods posed challenges.

The incorporation of GPS technology enabled the geo-location of detected road anomalies, facilitating the creation of pothole maps and timely repairs (Ramsha Suhail, 2020). (Desai et al., 2019) developed a process which identifies potholes and alerts the driver which creates cautiousness for the driver.

The advent of deep learning brought about a revolution, facilitating more reliable and precise detection of road anomalies (Kuthyar et al., 2021), (Rohitaa et al., 2021), (Paramarthalingam et al., 2024), (Silvester et al., 2019), (Chougule and Barhatte, 2023), (Ozoglu and Gökgöz, 2023). CNNs have been extensively used for image-based pothole and speed breaker detection (Rohitaa et al., 2021). (Shah and Deshmukh, 2019) classified road surfaces into potholes, speed bumps, and normal roads using a ResNet-50 CNN, achieving an 88.9% true positive rate. (Asad et al., 2022) classified pothole images under varying conditions using deep learning architectures like Tiny-YOLOv, YOLOv4, and YOLOv5, achieving accuracies of 80.04%, 85.48%, and 95%, respectively. In 2023, (Nissimagoudar et al., 2024) used YOLOv5 algorithm to detect potholes and speed breakers and claimed their approach can be useful for designing suspension systems in vehicles.

2.5 Crowdsourced and IoT-Based Solutions: Scaling Up Detection

Mobile sensing and crowdsourcing gained prominence as cost-effective means of gathering road condition data (Chuang et al., 2019). An Android application was created to detect road quality using data from triaxial accelerometers and gyroscopes (Mednis et al., 2011). Crowd sensing was used for the early detection of bumps and speed breakers, proving to be a cost-effective approach (Chuang et al., 2019). (Rahman et al., 2021) helped in identification of potholes using federated learning which produces notifications to drivers while encountering potholes and also updates the central server regarding updated road conditions.

ADAS now integrate road anomaly detection to alert drivers (-López et al., 2023). (Varma et al., 2018) proposed a method using deep learning to detect and inform drivers about unmarked and marked speed bumps/humps in real time. (Wang et al., 2015) proposed a real-time pothole detection approach to improve traffic safety for ITS. The technology alerts drivers in advance, advising them to slow down near GPS spots known for road abnormalities (Ramsha Suhail, 2020).

Characterizing speed bumps has become a focal point, with researchers aiming to identify different types based on their profiles. Deep learning models are used for detection and to characterize bumps into types such as sinusoidal, flat-top, thump, and round-top (Salman and Mian, 2023). Two-stage machine learning models have been introduced to enhance anomaly detection accuracy where a) Random Forest Classifier identifies potential anomalies in the first stage, while b) Gaussian Process Classifier provides detailed classification in the second stage

(Zhao et al., 2024). Adaptive systems are being developed to adjust detection algorithms based on vehicle speed and environmental conditions (Billah and Chattaraj, 2021).

In conclusion, the field has transitioned from basic sensor-based detection to sophisticated systems that use deep learning, GPS, and data fusion for more accurate and reliable road anomaly detection.

2.6 Summary

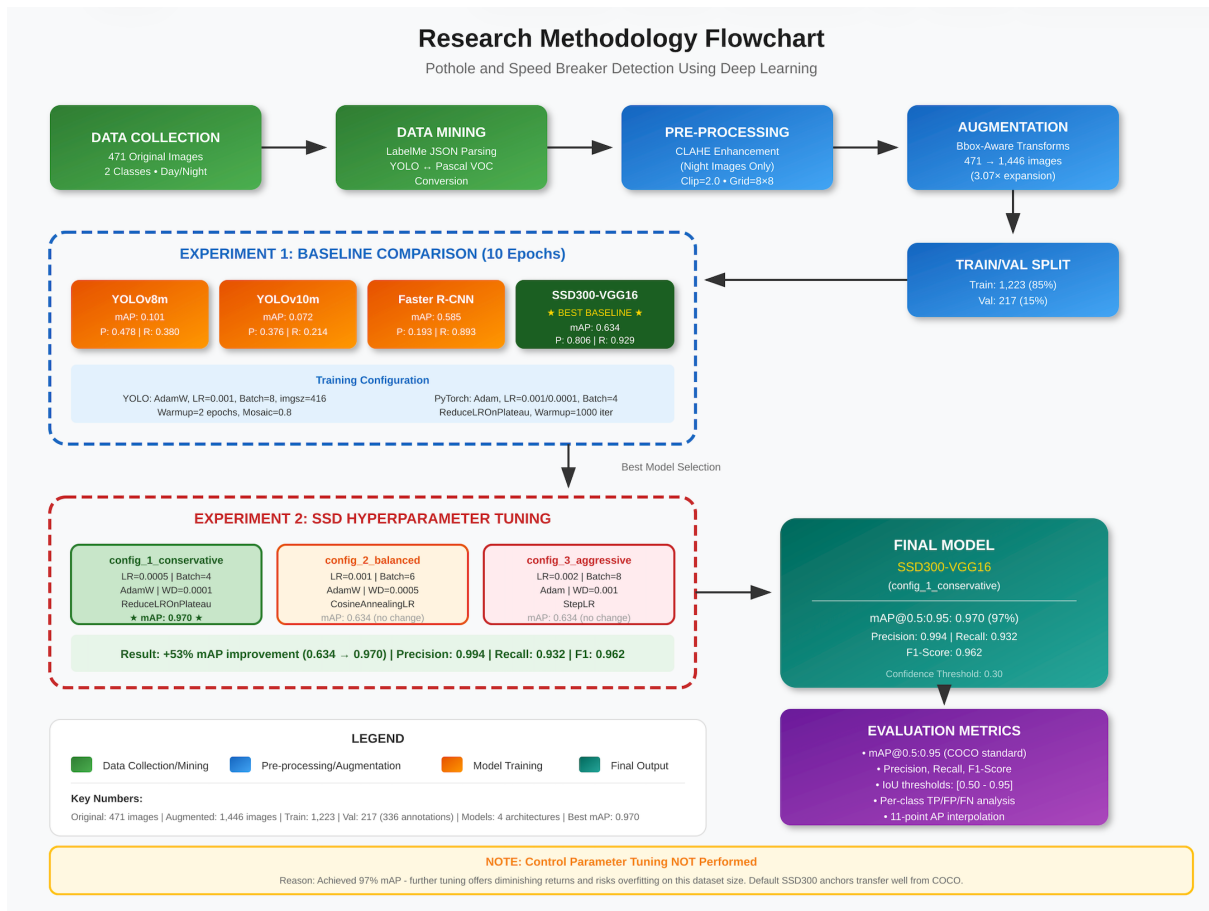
The related work section illustrates a clear evolution in road anomaly detection which was earlier a manual detection to advanced machine learning & deep learning detection along with infusion of GPS data to capture these road anomalies in real-time. The trend indicates a move towards more accurate, reliable, and real-time detection capabilities, with a significant emphasis on the application of deep learning models and their integration into ADAS and smart city initiatives

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This research aims to develop and evaluate robust machine learning algorithms for the accurate identification of potholes and speed breakers in diverse environmental conditions, using computer vision techniques – YOLO v8, YOLO v10 models, Faster R-CNN model, SSD model. To achieve this, the research follows a structured approach, it covers (i) data organization and pre-processing tailored for varying lighting and weather, (ii) label parsing and conversion, (iii) augmentation with bounding-box preservation, (iv) multiple detector families (YOLO, Faster R-CNN, SSD), (v) optimization and hyperparameter tuning, and (vi) rigorous evaluation using mAP, precision, recall and F1. Multiple modelling techniques as mentioned above are tested, and the most effective one is selected i.e. SSD is selected as the best model.

Figure 1: Showing block diagram of research methodology



3.2 Dataset Description:

Dataset contains 450+ samples of road signs, potholes, speed breakers in different conditions sunny, rainy, night as well as at different angles as well, center, right, left corner of the road. The dataset contains images captured from 2 phones iPhone12 & Vivo. The 2 classes in consideration here are speed breaker & potholes. Some of the images are labelled images which can be helpful to train the data. The dataset contains also unmarked speed breakers & potholes as well.

Key characteristics include:

- Lighting conditions: Daytime (clear, cloudy, overcast) and nighttime imagery
- Capture devices: iPhone 12 and Vivo smartphones
- Viewpoint variations: Center, left corner, and right corner perspectives
- Target classes: 2 classes - Pothole (class 0) and Speed Breaker (class 1)
- Annotation format: LabelMe JSON with YOLO text label conversion
- Scene diversity: Urban roads, rural paths, marked and unmarked anomalies

Training and validation sets are used for model development and selection. A held-out test set is reserved for final, unbiased reporting only.

Data augmentation will be used to address the limited dataset, by zooming, flipping, rotating images, etc. (Heltin Genitha et al., 2022).

3.3 Data Mining

Images and their corresponding annotation files were programmatically collected from the project's mounted storage and organized into a structured directory format. The dataset was arranged under folders such as:

Directory Structure:

- For speed breakers- main_dataset/original/ speed_breakers / daytime and main_dataset/original/speed_breakers/ nighttime
- For potholes- main_dataset/original/ potholes/ daytime and main_dataset/original/ potholes/ nighttime

The dataset includes **LabelMe JSON** files containing shape coordinates and labels, which were primarily used during preprocessing and visualization. For model training, **YOLO text label**

files (in the format class x_center y_center width height) were used. These YOLO labels were automatically converted into Pascal VOC (Visual Object Classes) **coordinates** when required for other detection models. This conversion ensured consistency and maintained a single source of truth for all annotations while enabling compatibility across different model architectures.

3.4 Data Pre-Processing:

Pre-processing enhances image quality and normalizes inputs prior to learning. Typical operations include noise reduction, contrast enhancement especially in low-light images and standardized sizing so models receive consistent inputs (-López et al., 2023). A **lighting-aware procedure applied CLAHE** to night images to boost local contrast while limiting noise amplification.

Contrast-Limited Adaptive Histogram Equalization (CLAHE) was applied specifically to nighttime images to boost local contrast while limiting noise amplification. CLAHE operates on small regions (tiles) rather than the entire image, preventing over-amplification in already high-contrast areas.

- Clip Limit: 2.0 (controls contrast limiting)
- Tile Grid Size: 8×8 pixels
- Application: Nighttime images only (selective enhancement)

Input resizing is NOT performed during pre-processing. Instead, resizing is handled internally by each training framework via built-in image-size parameters. This ensures that images and bounding boxes are scaled together atomically, preserving label fidelity and preventing annotation misalignment.

- YOLO models: imgsz=416 (internal resize)
- Faster R-CNN: Variable input with FPN handling
- SSD300: 300×300 fixed input (internal resize)

3.5 Data Transformation and Augmentation:

Augmentation increases the variety of the dataset to make the model more robust. Typical changes include brightness/contrast shifts, slight color adjustments, adding gentle noise or blur, and other small variations so the model can handle different conditions. To increase the size and variability of the dataset, data augmentation techniques are employed (Javed et al., 2021). The guiding principle was to introduce realistic variations that a road-facing camera would plausibly encounter (changes in brightness and colour, slight sensor noise, motion-related blur),

while preserving the true content of each scene. To ensure that these transformations were credible, sample visualizations of original and augmented images were inspected to confirm that (i) the visual appearance remained realistic and (ii) object annotations still matched the visible targets.

Bounding-box-aware augmentation was used so that images and labels stay in sync. Operations included modest adjustments to **brightness/contrast, color, noise, motion blur, and sharpening**. Night images received relatively more augmentation to balance conditions in the dataset. After each augmentation run, sample before/after panels were generated and reviewed to verify the authenticity of the transformations and the correct placement of bounding boxes. The following photometric and mild geometric transforms were applied:

- Brightness adjustment: $\pm 20\%$ variation
- Contrast adjustment: $\pm 15\%$ variation
- Color jitter: Hue, saturation shifts within $\pm 10\%$
- Gaussian noise: $\sigma = 0.01-0.03$ (sensor noise simulation)
- Motion blur: Kernel size 3-7 (vehicle movement simulation)
- Sharpening: Enhancement factor 0.5-1.5

As a direct outcome, the number of training images increased (because each augmented variant is saved as an additional sample). The study also tallied the counts before and after augmentation and summarized the distribution by class and lighting condition.

Table 1: Showing images count before & after image augmentation

Category	Before Augmentation	After Augmentation
Potholes - Daytime	361	1,073
Potholes - Nighttime	13	78
Speed Breakers - Daytime	95	283
Speed Breakers - Nighttime	2	12
TOTAL	471	1,446

3.6 Modelling Techniques:

Convolution Neural Networks (CNNs) are effective in extracting features from images and have demonstrated high accuracy in road anomaly detection (Bhandari et al., 2024), (Billah and Chattaraj, 2021). Architectures like VGG16, ResNet-50, and EfficientNet are evaluated for feature extraction and classification of road anomalies (-López et al., 2023), (Billah and Chattaraj, 2021).

Object detection models like YOLOv5, YOLOv7, YOLOv8, faster R-CNN are also used to precisely locate potholes & speed breakers (Lan et al., 2024). YOLOv5 is known for real-time performance with minimal computational overhead while Faster R-CNN provides higher accuracy but requires more processing power (Lieskovská et al., 2023).

This study evaluates one-stage and two-stage object detectors. One-stage models (e.g., YOLO, SSD) predict bounding boxes and classes in a single pass and are well suited to real-time applications. Two-stage models (e.g., Faster R-CNN) first propose candidate regions and then classify/refine them, often achieving higher accuracy at the cost of speed - (Lan et al., 2024), (Lieskovská et al., 2023).

Three representative detector families were trained and compared: YOLO, Faster R-CNN (with a lightweight backbone), and SSD. Thresholds for detection and non-maximum suppression were adjusted conservatively to reduce spurious detections while maintaining recall. Listing details on detector families below:

YOLO Family (YOLOv8m, YOLOv10m):

YOLO (You Only Look Once) models process entire images in a single evaluation, making them extremely fast for real-time applications. Two variants were evaluated:

- YOLOv8m: Medium variant with 25.84M parameters, 78.7 GFLOPs
- YOLOv10m: Enhanced variant with 15.31M parameters, 58.9 GFLOPs
- Backbone: CSPDarknet with Feature Pyramid Network (FPN)
- Head: Anchor-free detection with decoupled head

Faster R-CNN (MobileNetV3-FPN)

Faster R-CNN is a two-stage detector using Region Proposal Networks (RPN) for candidate generation followed by region-based classification:

- Backbone: MobileNetV3-Large with Feature Pyramid Network

- Parameters: ~70 total (56 backbone + 14 detector head)
- ROI Pooling: 7×7 aligned pooling
- Classification: FastRCNNPredictor with 3 classes (background + 2)

SSD300-VGG16

Single Shot MultiBox Detector (SSD) uses multi-scale feature maps for detection, balancing speed and accuracy:

- Backbone: VGG16 pre-trained on ImageNet
- Input Resolution: 300×300 pixels (fixed)
- Feature Maps: 6 scales (38×38 , 19×19 , 10×10 , 5×5 , 3×3 , 1×1)
- Default Boxes: Aspect ratios [1, 2, 3, 1/2, 1/3] per location
- Classification: Modified head for 3 classes

3.7 Model Training & Experiments:

3.7.1 Training Protocol:

All models were trained on the training split and monitored on the validation split. **Transfer learning** and **fine-tuning** were used. Checkpoints of the best validation performance were saved. **Input resizing was performed inside the training frameworks** via built-in image-size settings ensuring that images and bounding boxes were scaled together, which is why no resizing appears in pre-processing.

3.7.2 Experiment 1: Baseline Comparison

Objective. Establish a fair baseline across detector families and identify the main failure modes, especially at night.

Design. One representative from each family (YOLO, SSD, Faster R-CNN) was trained for 10 epochs under stable settings on the augmented training set and validated consistently. Thresholds were chosen to balance precision and recall without deep tuning.

YOLO Training Configuration:

- Epochs: 10
- Image Size: 416×416 pixels
- Batch Size: 8 (GPU) / 4 (CPU)
- Optimizer: AdamW
- Initial Learning Rate: 0.001
- Final Learning Rate Factor: 0.01
- Momentum: 0.937
- Weight Decay: 0.0005
- Warmup Epochs: 2.0
- Warmup Momentum: 0.8
- Warmup Bias LR: 0.1
- Box Loss Weight: 7.5
- Classification Loss Weight: 0.5
- Mosaic Augmentation: 0.8
- Mixup: 0.0
- Seed: 42 (reproducibility)

Faster R-CNN / SSD Training Configuration:

- Epochs: 10
- Batch Size: 4
- Optimizer: Adam
- Learning Rate (Backbone): 0.0001 (pre-trained, lower LR)
- Learning Rate (Detector Head): 0.001 (randomly initialized)
- Weight Decay: 0.0005
- Warmup: First epoch with linear warmup over 1000 iterations
- Warmup Factor: 0.001
- LR Scheduler: ReduceLROnPlateau (factor=0.5, patience=2, min_lr=1e-6)
- Gradient Clipping: max_norm=10.0
- Loss Clamping: Maximum loss capped at 100 for stability
- Confidence Threshold (Validation): 0.05

Evaluation: Metrics included **Precision**, **Recall**, **F1**, **IoU**, and **mAP** on the validation set. **Qualitative panels** (ground truth vs predictions) were assembled for **day** and **night** images to inspect typical successes/failures (e.g., reflective paint at night).

Findings (notes). Night scenes remained the most challenging; lighting enhancement and augmentation helped but did not eliminate the gap. Reflective markings introduced **false positives** for speed-breaker detections at night. Very small or oblique potholes were harder for single-pass detectors. **SSD** produced the **best validation accuracy at epoch 10** among the tested baselines while retaining reasonable efficiency; YOLO was fastest but slightly behind in this dataset setting; Faster R-CNN was more selective but slower.

Table 2: Showing evaluation metrics across computer vision models

Model	mAP@0.5:0.95	Precision	Recall	F1	Time (min)
YOLOv8m	0.101	0.478	0.380	0.424	--
YOLOv10m	0.072	0.376	0.214	0.273	--
Faster R-CNN (MBv3)	0.585	0.193	0.893	0.317	51.5
SSD300-VGG16	0.634	0.806	0.929	0.863	52.8

SSD peaked mid-training at mAP≈**0.746** but finalized at **0.634**; the **0.634** value is used as the tuning baseline.

Recommendations forwarded to Experiment 2 (explained).

- **Focus on SSD** as the lead architecture (it had the **best mAP at epoch 10** among baselines).
- **Extend training beyond epoch 10** using **early stopping** to capture further gains without overfitting.
- **Run a focused hyperparameter search** (optimizer, learning rate, weight decay, batch size, scheduler) around SSD to improve precision–recall balance, particularly at night.
- Keep **lighting-aware pre-processing** and **heavier night augmentation**; mildly adjust **confidence and NMS thresholds** to curb night-time false positives

3.7.3 Experiment 2: SSD Hyperparameter Tuning

Objective. Improve the SSD baseline’s precision-recall balance and overall mAP through limited but impactful tuning, while preserving efficiency.

Method. Training resumed from the epoch-10 SSD checkpoint. Three compact configurations were tested (up to ~30 additional epochs with early stopping and LR scheduling):

Table 3: Showing hyperparameter configuration for SSD

Configuration	LR	Batch	Weight Decay	Optimizer	Scheduler	Conf
config_1_conservative	0.0005	4	0.0001	AdamW	ReduceLROnPlateau	0.30
config_2_balanced	0.0010	6	0.0005	AdamW	CosineAnnealingLR	0.25
config_3_aggressive	0.0020	8	0.0010	Adam	StepLR	0.20

Validation Results - Baseline (epoch-10 SSD): mAP (baseline) ≈ 0.634 . Hyperparameter tuned results are as follows

- config_1_conservative: mAP = 0.970 (+0.336 improvement), training time ≈ 157.4 min $\rightarrow$ best.
- config_2_balanced: mAP = 0.634 (no improvement), ≈ 41.7 min.
- config_3_aggressive: mAP = 0.634 (no improvement), ≈ 36.2 min.

Conclusion from Experiment 2. The conservative regime (lower LR, smaller batch, mild WD, AdamW + ReduceLROnPlateau, conf $\approx$ 0.30) yields a substantial, stable improvement. Adopt SSD300-VGG16 with config_1_conservative as the study’s recommended detector (subject to test-set confirmation).

Recommendation from Experiment 2. Adopt SSD300 with config_1_conservative as the preferred model for deployment/next-stage evaluation. Retain the confidence threshold ($\sim$ 0.30) that yielded the best validation results, subject to verification on the held-out test set.

3.7.4 Control Parameter Tuning

Control parameter tuning (such as anchor box configurations, feature pyramid levels, or NMS thresholds) was NOT performed in this study. This decision was made for the following reasons:

- **Excellent Baseline Performance:** The SSD300-VGG16 model achieved 97% mAP@0.5:0.95 after hyperparameter tuning, which exceeds typical deployment requirements for road anomaly detection.
- **Diminishing Returns:** Given the already high accuracy (97%), further tuning of control parameters would likely yield marginal improvements (< 2-3%) while significantly increasing computational overhead.
- **Risk of Overfitting:** Extensive control parameter tuning on a dataset of this size (1,446 images) risks overfitting to dataset-specific characteristics rather than improving generalization.
- **Architectural Defaults Optimized:** SSD300's default anchor configurations and feature pyramid design are well-optimized from extensive research on COCO dataset and transfer effectively to road anomaly detection.
- **Future work may explore control parameter tuning if:** (a) dataset size increases significantly, (b) new challenging conditions are introduced, or (c) deployment constraints require specific precision-recall trade-offs.

3.8 Evaluation Metrics:

Formulation.

- **Precision (P)** = $TP/(TP+FP)$: positive predictive value.
- **Recall (R)** = $TP/(TP+FN)$: sensitivity.
- **F1** = $2PR/(P+R)$: harmonic mean of P and R.
- **IoU** = $|\hat{B} \cap B|/|\hat{B} \cup B|$; detections counted correct if $IoU \geq \tau$.
- **mAP@0.5:0.95**: mean AP across IoU thresholds in [0.50, 0.95].
Implementation note: Ultralytics reports standard COCO-style mAP; the custom evaluator for torchvision detectors averages AP over a subset of IoUs (e.g., 0.50 and 0.75), used consistently for **relative** model selection.

Observed change (SSD, validation).

- **mAP@0.5:0.95: 0.634 → 0.970 (+0.336 absolute).**
- **Precision: 0.806 → ≈0.994 (↑ +0.188).**
- **Recall: 0.929 → ≈0.932 (≈ stable).**
- **F1: 0.863 → ≈0.962 (↑ +0.099).**

Table 4: Showing hyperparameter tuning results for SSD

Configuration	mAP@0.5:0.95	Improvement	Training Time	Status
Baseline (Epoch 10)	0.634	--	52.8 min	Reference
config_1_conservative	0.970	+0.336 (+53%)	157.4 min	BEST
config_2_balanced	0.634	+0.000	41.7 min	No change
config_3_aggressive	0.634	+0.000	36.2 min	No change

Table 5: Showing hyperparameter tuning improvement for SSD

Metric	Baseline	After Tuning	Change
mAP@0.5:0.95	0.634	0.970	+0.336
Precision	0.806	0.994	+0.188
Recall	0.929	0.932	+0.003 (stable)
F1-Score	0.863	0.962	+0.099

3.9 Result Interpretation

The staged design structured curation, lighting-aware enhancement, bbox-safe augmentations, baseline comparison, and focused SSD tuning drives a **step-change in validation accuracy** while maintaining high recall. Moving from the SSD baseline (**mAP@0.5:0.95 = 0.634, P=0.806, R=0.929, F1=0.863**) to the tuned setting (**mAP=0.970, P≈0.994, R≈0.932, F1≈0.962**) demonstrates that optimization primarily **tightened precision** without sacrificing recall and improved overall localization quality. Residual error modes (nighttime reflections; very small/oblique potholes) motivate targeted data expansion (including **marked speed-breakers**) and, for deployment contexts with video, **temporal smoothing** or sequence models.

Final selection.

- **SSD300-VGG16 (config_1_conservative)** is the **recommended model**: **mAP@0.5:0.95 = 0.970** with **high precision and sustained recall** at the best-mAP epoch. Test-set confirmation and runtime/footprint analysis are reported separately in the Results/Discussion.

CHAPTER 4: RESULTS & DISCUSSION

4.1 Quantitative Results (Experiment 1)

Baseline validation outcomes: **SSD** final epoch mAP = 0.634 (peak ≈ 0.746), **Faster R-CNN** mAP = 0.585, **YOLOv8m** = 0.101, **YOLOv10m** = 0.072. Precision/Recall patterns indicate SSD's favorable balance at epoch 10.

4.2 Optimization Results (Experiment 2)

The **conservative SSD regime** improved validation **mAP to 0.970**, increased **precision**, and maintained **high recall**, particularly stabilizing night-scene detections.

4.3 Qualitative Analysis

GT vs prediction overlays confirm fewer **false speed-breaker positives** in night scenes post-tuning and tighter boxes for mid-scale potholes; very small/oblique anomalies remain difficult.

4.4 Error Analysis & Practical Implications

Persistent failure modes include **night glare**, **reflective paint**, and **tiny/oblique potholes**. For ADAS/ GIS (Geographic Information System) use, calibrate thresholds per condition; when mapping, apply conservative buffers around reported boxes.

CHAPTER 5: CONCLUSIONS & RECOMMENDATIONS

5.1 Conclusion:

This work presented supervised object-detection pipeline for speed-breaker and pothole detection across varied illumination and weather. The pipeline comprised lighting-aware pre-processing (CLAHE for night scenes), bbox-safe augmentation (photometric and mild geometric transforms), unified label handling (YOLO $\leftrightarrow$ VOC), and a comparative evaluation of YOLOv8/v10, Faster R-CNN (MobileNetV3-FPN), and SSD300-VGG16.

Experiment 1 (10-epoch baselines) established SSD300-VGG16 as the strongest performer on the validation set (final $\text{mAP}@0.5:0.95} \approx 0.634$, $P = 0.806$, $R = 0.929$, $F1 = 0.863$), outperforming YOLOv8/v10 and Faster R-CNN under the same data regime. Guided by these results, Experiment 2 resumed from the SSD epoch-10 checkpoint and executed a focused hyperparameter search. The conservative regime $\rightarrow$ AdamW, $LR = 5e-4$, $\text{batch} = 4$, $\text{weight-decay} = 1e-4$, ReduceLROnPlateau , $\text{conf} \approx 0.30$ yielded a substantial improvement to $\text{mAP}@0.5:0.95 = 0.970$ ($\Delta + 0.336$), with precision increasing to ≈ 0.994 at the best-mAP epoch while recall remained ≈ 0.932 ($F1 \approx 0.962$).

Despite the strong validation scores, error analysis indicates that high accuracy is driven by numerous visually similar images of the same scene from different angles. While this correlation aids optimization, it constrains cross-scene generalization and does not fully address difficult regimes (night glare, very small/oblique potholes). The principal methodological conclusion is that data scale and diversity rather than additional near-duplicates will most effectively improve box regression quality and robustness.

Final model recommendation. For this dataset and target latency, SSD300-VGG16 (config_1_conservative) is the preferred detector (subject to test-set confirmation and deployment profiling).

5.2 Future Work:

Although the tuned SSD model achieved high accuracy, further work can make the system more reliable and adaptable to real-world use.

1. Expand the dataset. The current results benefited from multiple images of similar scenes. Collecting a larger and more diverse dataset with different locations, lighting, weather, and camera devices will improve model generalization and bounding-box precision.
2. Improve model robustness. Future experiments should test domain adaptation so the model performs consistently across different road types and environments.
3. Optimize deployment. Techniques such as model pruning, quantization, and TensorRT conversion can make the SSD model lighter and faster for real-time or mobile use.

Extend to video input. Integrating temporal smoothing or short-sequence tracking could reduce flickering detections and make the model more stable for driving-camera applications.

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APPENDIX A: RESEARCH PLAN

Table 6: Showing Research Plan

ACTIVITY	PERIODS - Weeks																				
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Literature search																					
Literature review																					
Research Proposal Submission																					
Explore Data																					
Data Pre-Processing & Augmentation																					
Interim Report Submission																					
Develop & test Models																					
Hyperparameter Tuning																					
Analyse & Evaluate																					
Final Thesis Submission																					

Figure 1: Showing Legends of Gantt Chart



APPENDIX B: RESEARCH PROPOSAL

POTHOLE AND SPEED BREAKER DETECTION

AYUSH KUMAR

Research Proposal

FEBRUARY 2025

Abstract

This research addresses the critical need for reliable road anomaly detection, specifically targeting potholes and speed breakers, which present ongoing safety and economic challenges. Current methods often lack the robustness required for varied real-world conditions. This proposal outlines the development of robust machine learning algorithms, leveraging computer vision, to accurately identify these anomalies across diverse environmental conditions. Drawing on the strengths of CNNs highlighted in recent studies, this project will focus on creating models capable of extracting meaningful features from raw image data to make accurate predictions. The primary aim is to build and evaluate algorithms that overcome the limitations of existing systems, such as difficulties with the detection of speed bumps & potholes under adverse environmental conditions. The expected outcome is a machine learning-based solution demonstrating high accuracy, precision, and recall rates in real-world settings, ultimately enhancing the safety and autonomy of vehicles on the road.

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LIST OF ABBREVIATIONS

CNNs:	Convolutional Neural Networks
SRGANs:	Super-Resolution Generative Adversarial Networks
ADAS:	Autonomous Driving Assistant System
YOLOv8:	You Only Look Once version 8
YOLOv7:	You Only Look Once version 7
GLCM:	Gray-Level Co-Occurrence Matrix
IoU:	Intersection over Union
mAP:	Mean Average Precision
ML:	Machine Learning
DL:	Deep Learning

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3.1 Background

Potholes and improperly designed speed breakers pose significant challenges to road safety and infrastructure maintenance, particularly in countries like India. In 2022, potholes were responsible for over 4,400 accidents across the nation. (India: road accidents due to potholes | Statista, 2025). These hazardous road conditions not only lead to accidents but also escalate vehicle maintenance costs, fuel consumption, and travel times, imposing substantial economic burdens.

Speed breakers are intended to regulate vehicular speed, but can become perilous when poorly designed. Studies have shown that non-compliant speed breakers often lead to abrupt vehicle movements, loss of control, and increased wear and tear, making them hazardous. (Kumar et al., n.d.).

The economic ramifications of potholes and substandard speed breakers are profound. Damaged roads contribute to higher fuel consumption, increased vehicle repair costs, and disrupted logistics. Traffic congestion resulting from these defects further hampers transportation efficiency, adversely affecting both urban and rural economies. Addressing these challenges necessitates innovative approaches to road maintenance and safety.

Advancements in machine learning and computer vision offer promising solutions for detecting and assessing road anomalies. AI-driven systems can analyse images, sensor data, and real-time traffic information to identify potholes and speed breakers with high precision. For instance, recent research has demonstrated the effectiveness of deep learning models, such as YOLOv8, in accurately detecting road hazards, including potholes and manholes, under various conditions (Khare et al., 2023)

Additionally, the integration of Super-Resolution Generative Adversarial Networks (SRGANs) with object detection models like YOLOv7 has shown improved accuracy in identifying potholes, even in low-resolution images.

As urbanization intensifies and vehicle numbers surge, integrating intelligent road monitoring solutions into existing transportation frameworks becomes imperative. Machine learning-based detection methods offer a cost-effective and scalable approach to identifying hazardous road

conditions, ultimately enhancing safety, improving traffic flow, and preserving infrastructure integrity.

3.2 Problem Statement

Road anomalies like potholes and speed bumps present significant risks to road safety and can cause vehicle damage, underscoring the need for automated detection systems, particularly given the inefficiency of traditional methods like manual assessment. The accurate detection of these anomalies is complicated by several factors such as varying sizes and shapes of potholes, uneven speed breaker construction, lack of adherence to government norms in speed breaker design, and absence of proper markings or cautionary signs. Adverse environmental conditions, including rain, poor lighting, varying weather, distance, and speed, add to the challenge. Limitations in camera quality, sensor accuracy, and computational capabilities further impact the reliability of real-time detection, especially considering street images filled with redundant information, substantial noise, and contamination. To overcome these limitations, this research focuses on leveraging machine learning and computer vision algorithms to enhance detection accuracy and reliability, effectively detecting speed breakers and potholes from different camera angles as well as under diverse environmental conditions.

3.3 Related Work

Early research on road anomaly detection primarily utilized basic techniques such as threshold methods and manual analysis of sensor data. These initial approaches often involved visual inspections or simple vibration-based methods, which were limited by their inefficiency and subjectivity. The need for more efficient and automated systems became apparent as technology advanced.

2.1 Early Research and Sensor-Based Approaches

Early detection methods for road anomalies relied on **vibration-based systems**, primarily using **accelerometers**. (Vittorio et al., 2014) measured vehicle vibrations and compared them against predetermined thresholds to identify potential potholes or speed breakers. This approach provides real-time data about road conditions to facilitate road repairs. This method was cost-effective but contained high false-positive rates. (Mohan et al., 2008) utilized anomaly detection thresholds with accelerometer data. (Perttunen et al., 2011) applied machine learning algorithms to accelerometer data for improved accuracy. (Mednis et al., 2011) introduced a

real-time pothole detection system using Android smartphones and accelerometer data. (Bhoraskar et al., 2012) also used accelerometers to detect road anomalies. (Ramsha Suhail, 2020) proposed the use of piezo-electric sensors attached to the shock absorbers of the front wheels to detect voltage changes caused by road anomalies.

2.2 Transition to Image Processing Techniques

Image processing techniques then came into play, with researchers creating algorithms to spot potholes and speed breakers in road images (Devapriya et al., 2016). (He et al., 2011) introduced a model using two CCD cameras and LED linear light to detect the 3-dimensional cross-section of pavement potholes. (Devapriya et al., 2016) used Gaussian filtering and median filtering to eliminate noise in images to detect speed bumps.

2.3 Transition to Machine Learning & Deep Learning Approaches

The **integration of machine learning algorithms** marked a significant step toward improving the accuracy and efficiency of road anomaly detection (Gaisser and Jonker, 2017). (Jain et al., 2012) used machine learning for bump detection using accelerometers in various vehicles in 2012. SVM models were employed for pothole detection in pavement images (He et al., 2011). (Bharathi et al., n.d.) proposed a speed breaker identification method using gray-level co-occurrence matrix (GLCM) features. In 2023, (Billah and Chattaraj, 2021) proposed adaptively choosing deep learning models like ResNet50, EfficientNet B7, VGG16 and AlexNet using image processing to detect road conditions.

3D reconstruction methods, **leveraging 3D laser scanning and stereo vision**, emerged, enabling the creation of detailed road surface models (Kim et al., 2022). (Li et al., 2009) presented a real-time 3D laser scanning system for detecting pavement rutting, shoving, and potholes. (Hou et al., 2007) proposed a stereo-vision-based surface model for better pavement conditioning. (Chang et al., 2005) introduced a clustering approach to quantify the severity and coverage of potholes using 3D point clouds. However, the high costs and computational demands associated with these methods posed challenges

The incorporation of **GPS technology** enabled the geo-location of detected road anomalies, facilitating the creation of pothole maps and timely repairs (Ramsha Suhail, 2020). (Desai et al., 2019) developed a process which identifies potholes and alerts the driver which creates cautiousness for the driver.

The advent of **deep learning** brought about a revolution, facilitating more reliable and precise detection of road anomalies (Kuthyar et al., 2021), (Rohitaa et al., 2021), (Paramarthalingam et al., 2024), (Silvester et al., 2019), (Chougule and Barhatte, 2023), (Ozoglu and Gökgöz, 2023). CNNs have been extensively used for image-based pothole and speed breaker detection (Rohitaa et al., 2021). (Shah and Deshmukh, 2019) classified road surfaces into potholes, speed bumps, and normal roads using a ResNet-50 CNN, achieving an 88.9% true positive rate. (Asad et al., 2022) classified pothole images under varying conditions using deep learning architectures like Tiny-YOLOv, YOLOv4, and YOLOv5, achieving accuracies of 80.04%, 85.48%, and 95%, respectively. In 2023, (Nissimagoudar et al., 2024) used YOLOv5 algorithm to detect potholes and speed breakers and claimed their approach can be useful for designing suspension systems in vehicles.

Mobile sensing and crowdsourcing gained prominence as cost-effective means of gathering road condition data (Chuang et al., 2019). An Android application was created to detect road quality using data from triaxial accelerometers and gyroscopes (Mednis et al., 2011). Crowd sensing was used for the early detection of bumps and speed breakers, proving to be a cost-effective approach (Chuang et al., 2019).

ADAS now integrate road anomaly detection to alert drivers (-López et al., 2023). (Varma et al., 2018) proposed a method using deep learning to detect and inform drivers about unmarked and marked speed bumps/humps in real time. (Wang et al., 2015) proposed a real-time pothole detection approach to improve traffic safety for ITS. The technology alerts drivers in advance, advising them to slow down near GPS spots known for road abnormalities (Ramsha Suhail, 2020).

Characterizing speed bumps has become a focal point, with researchers aiming to identify different types based on their profiles. Deep learning models are used for detection and to characterize bumps into types such as sinusoidal, flat-top, thump, and round-top (Salman and Mian, 2023). **Two-stage machine learning** models have been introduced to enhance anomaly detection accuracy where a) Random Forest Classifier identifies potential anomalies in the first stage, while b) Gaussian Process Classifier provides detailed classification in the second stage (Zhao et al., 2024). **Adaptive systems** are being developed to adjust detection algorithms based on vehicle speed and environmental conditions (Billah and Chattaraj, 2021).

In conclusion, the field has transitioned from basic sensor-based detection to sophisticated systems that use deep learning, GPS, and data fusion for more accurate and reliable road anomaly detection.

3.4 Aims & Objectives

The main aim of this research is to develop a robust & efficient model for the detection of speed breakers & potholes using machine learning & computer vision techniques, ensuring reliable performance across various weather conditions and lighting scenarios. The goal of this research is to contribute to effective speed breaker & pothole detection which can be utilized as a robust model in ADAS (Autonomous Driving Assistant System) or for GPS marking of potholes & speed-breakers.

The research objectives are formulated based on the aim of the study, which are as follows:

- To implement pre-processing techniques to enhance image quality, reduce noise & handle different lighting/ weather conditions.
- To design and implement machine learning models - CNNs, RNNs, and other deep learning architectures, for accurate detection and classification of speed breakers and potholes.
- To evaluate performance under various adverse environmental conditions (varying lighting, weather, and road surface).
- To identify the optimal ML/DL model(s) that provide the best balance of accuracy, recall & precision.

3.5 Significance of the Study

This research has the potential to greatly contribute to safer roads & lesser injuries/ fatalities to the public with economic benefits at national and international levels. Key aspects highlighting the importance and potential implications of this study include:

- **Reducing Accidents:** Systems detecting potholes and unmarked speed bumps can give drivers timely warnings so they can avoid dangers, which lowers the number of accidents and injuries.
- **Keeping Vehicles in Good Repair:** Systems enabling drivers to avoid road flaws can lower the cost of vehicle repairs and extend the life of the vehicle.
- **Autonomous Vehicle Advancement:** Autonomous Driving Assistant System (ADAS) cars can accurately detect road anomalies to be safer and more reliable.
- **Improved Road Safety:** Deployment on a large-scale the system has the potential to monitor road conditions and alert drivers of potential hazards, which can reduce accidents and injuries.

- **Economic Advantages:** Lowering vehicle maintenance expenses, enhancing road maintenance effectiveness, and cutting down on accidents can result in considerable financial gains for people, businesses, and governments.
- **Worldwide Applicability:** The research techniques and concepts can be used on road networks around the world, which will improve transportation systems' safety and efficiency.
- **Smart City Initiatives:** Integrating pothole and speed bump detection systems into smart city projects can give city officials and residents real-time data on road conditions.
- **Technology to Assist:** By offering audio alerts regarding road conditions, the technology can be tailored to help people who are blind.
- **Guiding Policy Makers:** The study can help governments and decision-makers spot unauthorized or poorly constructed speed bumps, which will lower the number of accidents and the related financial losses brought on by subpar road infrastructure.

3.6 Scope of the Study

There are following items which are **in-scope** under this research plan.

- **Image Data:** Focus on **identifying** these road anomalies using image data.
- **Adverse Environmental Conditions:** Specifically address detection challenges posed by varying lighting, weather (rain, fog), and road surface conditions.
- **Developing Machine Learning (ML) and Deep Learning (DL) Models:** Development, comparison, and evaluation of different ML/DL models.
- **Performance Metrics:** Evaluation using metrics like accuracy, precision, recall, F1-score, and processing time.

There are following items which are **Out-of-Scope** under this research plan.

- **Sensor Data:** Excluding the integration of data from other sensors such as accelerometers, LiDAR, or radar
- **GPS Integration:** Excluding the use of GPS data for location mapping or real-time driver alerts.
- **Real-Time Implementation:** Avoiding real-time implementation in a vehicle or mobile device.
- **3D Reconstruction:** Excluding **3D pavement reconstruction methods** that employ laser scanning or stereo vision.

- **Vibration-Based Methods:** Not using vibration-based approaches that rely on acceleration sensors.
- **ADAS or Autonomous Vehicle Integration:** Not integrating the system directly into Advanced Driver Assistance Systems (ADAS) or autonomous vehicle platforms.
- **Implementation on specific devices:** Not implementing the model on a mobile phone

3.7 Research Methodology

This research aims to develop and evaluate robust machine learning algorithms for the accurate identification of potholes and speed breakers in diverse environmental conditions, using computer vision techniques. To achieve this, the research follows a structured approach, encompassing data collection, pre-processing, feature engineering, model development, and evaluation. Multiple modelling techniques will be tested, and the most effective one will be selected based on empirical performance.

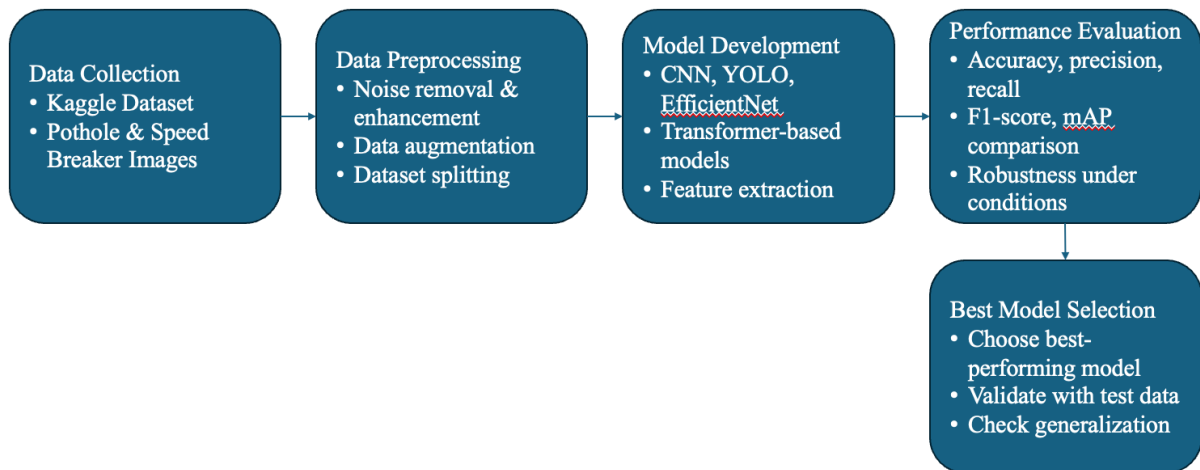


Figure 2: Showing block diagram of research methodology

3.1 Dataset Description:

Dataset contains 850+ samples of road signs, potholes, speed breakers in different conditions sunny, rainy, night as well as at different angles as well, center, right, left corner of the road. The dataset contains images captured from 2 phones iPhone12 & Vivo. Some of the images are labelled images which can be helpful to train the data. However, the dataset contains unmarked speed breakers only, further images containing marked speed breakers are to be added here. The dataset will be split into three distinct subsets: training, validation, and testing. Data augmentation will be used to address the limited dataset, by zooming, flipping, rotating images, etc. (Heltin Genitha et al., 2022).

3.2 Data Pre-Processing:

Image enhancement is performed to enhance the quality & clarity of the images. Techniques such as Gaussian filtering and median filtering is applied to remove noise as used by (-López et al., 2023). Noise from various sources is removed using filters. The images are resized to ensure uniformity and compatibility with the chosen machine-learning models. The images are converted to grayscale to reduce computational complexity and focus on essential features.

3.3 Data Transformation and Augmentation:

To increase the size and variability of the dataset, data augmentation techniques are employed (Javed et al., 2021). This includes:

- Rotation: Rotating images to simulate different camera angles.
- Scaling: Zooming in and out to represent objects at various distances.
- Translation: Shifting images horizontally and vertically.
- Flipping: Horizontally or vertically flipping images.

3.4 Modelling Techniques:

Convolution Neural Networks (CNNs) are effective in extracting features from images and have demonstrated high accuracy in road anomaly detection (Bhandari et al., 2024), (Billah and Chattaraj, 2021). Architectures like **VGG16**, **ResNet-50**, and **EfficientNet** are evaluated for feature extraction and classification of road anomalies (-López et al., 2023), (Billah and Chattaraj, 2021).

Object detection models like YOLOv5, YOLOv7, YOLOv8, faster R-CNN are also used to precisely locate potholes & speed breakers (Lan et al., 2024). YOLOv5 is known for real-time performance with minimal computational overhead while Faster R-CNN provides higher accuracy but requires more processing power (Lieskovská et al., 2023).

3.5 Model Training and Optimization:

Each model is trained using **transfer learning** and **fine-tuning techniques** to accelerate convergence and improve generalization. Key aspects include:

- Optimization Algorithms: Different optimization algorithms are used like Adam & SGD.

- **Hyperparameter tuning:** Further hyperparameter tuning is performed to attain better accuracy for the models. Parameters like learning rate, batch size, dropout rates, the number of convolution filters, size of convolution filters, and number of neural network outputs are tuned. (-López et al., 2023).

3.6 Evaluation Metrics:

To determine the best-performing model, the following metrics are analyzed:

- **Precision, Recall, and F1-score:** Assess model accuracy in classifying road anomalies.
- **Intersection over Union (IoU):** Evaluates object detection accuracy.
- **Mean Average Precision (mAP):** Measures object detection performance across varying confidence thresholds.
- **Inference Time & Model Size:** Determines real-time feasibility for deployment in edge computing and mobile devices.

After evaluation, the best-performing model is selected for real-world implementation. By focusing on these steps, this research provides valuable insights into developing effective and reliable machine learning algorithms for road anomaly detection, contributing to safer and more efficient transportation systems.

3.8 Required Resources

3.8.1 Hardware Requirements:

- **Camera:** A camera to capture additional road images (if needed)
- **High-Performance GPU:** Google Colab Pro, **NVIDIA RTX 3090 / A100 / Tesla V100** – Required for deep learning model training and inference acceleration.
- **CPU:** Intel Core i9 (12th Gen) / AMD Ryzen 9 (5900X or higher) – Ensures efficient data pre-processing and computational performance.
- **RAM: 32GB or higher** – Required for handling large datasets and multi-threaded processing.
- **Storage:** 2TB SSD + 4TB HDD – SSD for faster data access and HDD for long-term dataset storage.

3.8.2 Software Requirements: Mention OPENCV, pytorch

- **Operating System:** Mac OS
- **Coding Language:** Python. For example, Python 3.8.64.
- **Pre-Processing Framework:** Python Open CV.
- **Annotation Tools:** Roboflow/ labelme (if required) for labelling images.
- **Deep Learning Frameworks:** Such as TensorFlow 2.0 and Keras 2.4.04, Pytorch.
- **Development Environment:** Jupyter using Anaconda Framework.

3.9 Research Plan

Table 7: Block Diagram Showing Research Methodology

ACTIVITY	PERIODS - Weeks																				
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Literature search																					
Literature review																					
Research Proposal Submission																					
Explore Data																					
Data Pre-Processing & Augmentation																					
Interim Report Submission																					
Develop & test Models																					
Hyperparameter Tuning																					
Analyse & Evaluate																					
Final Thesis Submission																					



Figure 3: Showing Legends of Gantt Chart

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